SUPPORT-AWARE OPENING-OFFER OPTIMIZATION

Support-Aware Optimization of Buyer Opening Offers

Final research design, implementation audit, and full-data results

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Research question: How can a buyer's opening-offer ratio in eBay Best Offer negotiations be chosen to maximize model-estimated immediate expected savings while remaining within historical support?

5.71M	valid fashion opening offers
0.834	test ROC AUC
70%	simple support-safe default
17/17	automated tests passing

Objective

P(opening offer immediately accepted | state, anchor) x (1 - anchor). This is a one-step contextual-bandit objective and not an estimate of resale value, total buyer welfare, or the causal psychological effect of anchoring.

Executive summary

> **Bottom line: a 70% opening offer is the simplest defensible default. The personalized supervised policy predicts more savings, but CQL is substantially more support-conservative and does not beat the fixed rule on predicted reward.**

- The corrected unit is the first event in each buyer-item thread, retained only when it is a buyer opening offer.
- Immediate acceptance is opening-row status 1 or 9; status 2 is empirically inconsistent with an accepted offer.
- The acquisition price is the accepted offer itself, so immediate savings is 1 - anchor ratio.
- Listing-grouped train/validation/test splits have zero item overlap.
- H1 is supported, H2 is partially supported, and H3 is supported only as simulator sensitivity.

Key full-data results

9.73%	observed historical immediate savings
12.68%	fixed 0.70 model estimate
13.71%	supervised greedy model estimate
12.19%	CQL model estimate

Academic contribution

The contribution is not a claim that reinforcement learning automatically wins. It is a reproducible, evidence-separated comparison of calibrated supervised optimization, support-conservative offline learning, and simulator-only PPO. The useful findings are the interior optimum, the reward-support trade-off, and the measured shrinkage of PPO reward under robustness checks.

Decision boundary

Historical outcomes, Phase-1 model estimates, OPE diagnostics, and PPO simulator rewards are different evidence types. They are shown together for interpretation, not as a causal leaderboard.

1. Corrected research design

Observation and label

A bargaining thread is (anon_item_id, anon_byr_id). Events are ordered by src_cre_date. A thread enters the sample only when its first event has offr_type_id = 0. The target is the status of that same opening row: statuses 1 and 9 are accepted. A later deal does not retroactively label the opening offer as accepted.

Reward and scope

The action is offr_price / start_price_usd, restricted to 0.01 < anchor <= 1.00. Reward equals opening_accepted x (1 - anchor). Rejected, expired, and countered openings receive zero immediate reward. The default filter restricts the analysis to fashion meta-category 11450.

Status-code adjudication

Status	Rows	Price present	Median item/offer	Within 1%	Interpretation
1	1,754,643	99.99%	1.00	100.00%	Accepted
9	158,060	100.00%	1.00	100.00%	Accepted
2	743,683	56.02%	1.54	1.12%	Not accepted
6	1,161,592	58.07%	1.43	0.51%	Not accepted
7	1,549,864	65.33%	1.30	2.50%	Not accepted

Why the old reward failed: item_price tracks a reference/listing value for many non-accepted states. For accepted statuses 1 and 9, offr_price is the realized transaction price and directly implies the buyer's discount.

Leakage-safe sample

Split	Rows	Unique listings	Acceptance	Mean anchor	Item overlap
Train	4,564,662	3,273,530	33.52%	60.64%	0
Validation	570,404	409,191	33.57%	60.65%	0
Test	570,827	409,192	33.46%	60.62%	0

2. Models and outputs

State features

Feature	Meaning	Practical differentiation
log_list_price	Log listing price	Lower- and higher-priced listings
seller_score_norm	Normalized feedback volume	Established vs. new sellers
seller_pos_pct	Positive-feedback percentage	Seller reputation
categ_id_clean	Top 25 fashion leaf categories + other	Fashion subcategory

The recommender returns a ratio, dollar offer, predicted immediate acceptance, expected immediate savings, and support flag. It does not know brand, condition, material, demand, buyer valuation, or resale price.

Pipeline

Component	Implementation	Role / evidence
Historical	Observed opening outcomes	Behavior benchmark
Phase 1	XGBoost acceptance classifier	Calibrated response surface
Fixed rule	Anchor = 0.70	Simple support-safe baseline
Greedy	Grid maximizer of P(accept) x discount	Personalized supervised policy
CQL	Terminal one-step target; gamma=0; alpha=5	Conservative offline policy
PPO	Beta actor in learned simulator	Simulator sensitivity only
OPE	Kernel IPS/SNIPS/DR with estimated behavior	Support diagnostic, not causal

Implementation corrections

- Phase 1 artifacts are now created before downstream phases.
- CQL saves its scaler, history, metrics, and model; every transition is terminal.
- PPO sends raw engineered features to XGBoost and normalized states only to the actor.
- The PPO actor uses a bounded Beta distribution, eliminating clipped-Gaussian density bias.
- Flat orchestration replaces phantom module imports, and preprocessing is active.

3. Full-data results

0.8339	classifier ROC AUC
0.1539	classifier Brier score
0.4660	classifier log loss
33.46%	test opening acceptance

Policy	Evidence	Anchor	Accept	Savings	Support
Historical behavior	Observed	0.6062	0.3346	0.0973	1.0000
Fixed 0.70	Phase-1	0.7000	0.4226	0.1268	1.0000
Supervised greedy	Phase-1	0.7333	0.5007	0.1371	0.9099
One-step CQL	Phase-1	0.7025	0.4214	0.1219	0.9992
PPO basic	Simulator	0.7455	0.4823	0.1273	n/a
PPO robust	Simulator	0.7359	0.4515	0.1209	n/a

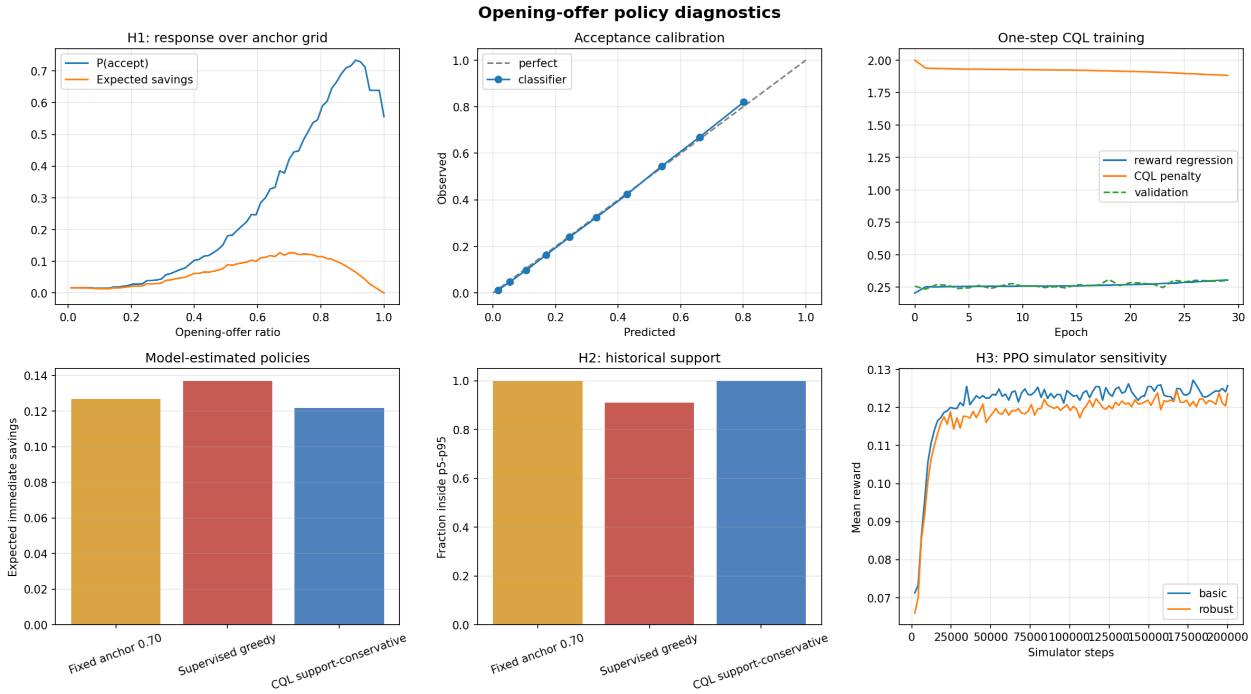
Do not read this table as causal lift. Historical savings is observed; fixed, greedy, and CQL values use the Phase-1 model; PPO values exist only inside that learned simulator.

OPE diagnostic at bandwidth 0.05

Policy	Reward model	SNIPS (95% interval)	DR (95% interval)	ESS
Fixed 0.70	0.1270	0.1257 (0.1240-0.1282)	0.1274 (0.1254-0.1302)	35.5%
Supervised greedy	0.1373	0.1277 (0.1257-0.1299)	0.1393 (0.1369-0.1419)	25.6%
CQL	0.1221	0.1235 (0.1216-0.1252)	0.1232 (0.1211-0.1252)	36.2%

Propensities are estimated, and the results vary with kernel bandwidth. The OPE table is a support diagnostic, not a causal policy evaluation.

4. Diagnostic dashboard



Top row: the response curve has an interior expected-savings maximum; classifier calibration follows the diagonal; CQL training remains stable. Bottom row: greedy predicts the largest savings but leaves central support more often; CQL remains almost entirely in support; robust PPO reward stays below basic PPO reward.

5. Hypothesis assessment

Hypothesis	Decision	Evidence
H1: interior optimum	SUPPORTED	99.981% of greedy actions are interior; population curve peaks near 0.67.
H2: CQL reduces support/extrapolation gap	PARTIAL	Greedy support 90.99% vs CQL 99.92%; model-SNIPS gap is smaller for CQL, but OPE is bandwidth-sensitive and non-causal.
H3: PPO gain shrinks under robustness	SUPPORTED IN SIMULATOR	Expected reward falls from 0.1273 to 0.1209, a 4.99% relative reduction.

Interpretation

The study's strongest result is not that an RL method dominates. The fixed 70% baseline is simple, fully supported, and has higher Phase-1 predicted savings than CQL. Greedy gains model-estimated reward but incurs more extrapolation. CQL buys support at a measurable reward cost. PPO's apparent advantage contracts under a modest simulator stress test.

What changed relative to the original research question

The word 'optimal' is narrowed to model-estimated immediate expected savings. The reward no longer uses item_price, and the outcome is immediate opening acceptance rather than eventual deal completion. The project can recommend an opening bid; it cannot claim to maximize resale margin or total buyer welfare.

Academic contribution retained

Yes. The contribution remains a rigorous comparison of methods under observational support constraints. A negative result for CQL is academically useful because it identifies when additional algorithmic complexity is not justified by the available data.

6. Practical use

Recommended default: start near 70% of listing price. Use personalized greedy advice only when its support flag is true; otherwise use CQL or the fixed rule.

Use case	Policy	Why
Fast rule without software	Fixed 70%	Simple, central-support action; 12.68% model-estimated expected savings.
Personalized in-support listing	Supervised greedy	Uses price, seller reputation, and leaf category; highest model estimate.
Greedy action flagged extrapolated	CQL or fixed 70%	Avoids unsupported optimizer recommendations.
Research stress test	Basic vs robust PPO	Measures simulator sensitivity; not deployment evidence.

Example output

Field	Meaning
recommended_anchor	Suggested opening offer divided by listing price
recommended_offer_usd	Anchor multiplied by the listing price
predicted_p_accept	Model-estimated immediate acceptance probability
predicted_expected_savings	Probability x discount; not guaranteed savings
within_support	Whether the recommendation lies inside historical p5-p95

Important qualification

A buyer who highly values obtaining the item may rationally choose a higher offer than this savings-only objective recommends. A reseller would need an independent resale-value estimate; this dataset does not observe the later sale of the same physical item.

7. Limitations and reproducibility

Limitations

- Countered, expired, and declined openings all receive zero, ignoring the option value of a seller counteroffer.
- Actions are endogenous. Unobserved quality, perceived overpricing, buyer information, and seller reservation values can confound the response surface.
- Brand, condition, material, demand, detailed buyer history, and resale prices are absent from the state.
- Fashion-only results may not generalize to other categories or time periods.
- PPO inherits the Phase-1 model's errors and is not evidence of live-marketplace performance.

Reproducibility checks

- Full run from 47.38 million source events to 5.71 million valid opening offers.
- Zero listing overlap across train, validation, and test.
- Saved status, anchor-response, calibration, support, OPE, and training diagnostics.
- CQL records gamma=0, alpha=5, and ten negative actions.
- Both PPO variants use the same seed and bounded Beta action representation.
- Seventeen of seventeen automated tests pass.

References

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Final conclusion: the corrected project supports a practical 70% default, a support-aware personalized option, and a defensible academic story without claiming causal lift or unobserved resale profit.